

Action Recognition and Temporal Action Segmentation Problems - Project report

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Abstract—This report presents the details about the approaches used to solve two machine learning problems: Action Recognition and Temporal Action Segmentation. For Action Recognition, a stacked ensemble of bidirectional Gated Recurrent Units (BiGRU), Random Forests, and Histogram-based Gradient Boosting Classifiers (BiGRU + RF + HGB) is implemented. For Temporal Action Segmentation, a deep sequence-to-sequence BiGRU trained on raw and differential features is used, combined with a temporal post-processing filter to address over-segmentation.

Index Terms—Robotic Assembly, Action Recognition, Temporal Action Segmentation, BiGRU.

I. INTRODUCTION

In order to train the models, the subset of the REASSEMBLE dataset [1] is provided containing 4 primary actions across 3 distinct objects, making 13 unique categories. It also contains high-frequency multimodal streams capturing robot proprioception, external forces, and visual frames.

Two primary problems are:

- 1) **Action Recognition**: Classifying pre-trimmed, variable-length demonstration clips into one of the 13 categories.
- 2) **Temporal Action Segmentation**: Processing an untrimmed stream of continuous observations to perform dense, frame-level action labeling.

The project dataset splits consist of 98 training demonstrations and 18 validation demonstrations. The remaining 21 demonstrations are withheld for test-set evaluation.

II. ACTION RECOGNITION APPROACH

A. Feature Engineering

RGB frames were bypassed to avoid GPU memory exhaustion on ≈ 200 -frame clips. All information derives from the seven proprioceptive signals. Two feature branches are computed per clip:

- Temporal sequence (63 dims/frame). The 37-dim proprioception vector is augmented with first- and second-order finite differences of pose and wrench, giving the recurrent model direct access to joint dynamics without requiring it to gain derivatives from raw values.
- Global statistics vector (71 dims/clip). Hand-engineered features targeting the two orthogonal classification dimensions of the task, like gear size (minimum gripper width, terminal grip width, joint configuration and end-effector

XYZ at the tightest-grip frame), action type (peak and mean wrench, gripper delta, mean linear velocity) and phase features (mean gripper width, mean wrench, mean XYZ during the action moment).

B. Model Architecture

A single-layer bidirectional GRU (hidden size 96, $\approx 135k$ parameters) processes the temporal sequence. Mean and max pooling over time (384 dims) are fused with a two-layer MLP applied to the global statistics (32 dims). The final classifier maps the 416-dim vector through a hidden layer (64 units) to the 13 class logits. Model capacity was deliberately kept lean: an early 742k-parameter variant regressed to below 70% validation F1 due to overfitting on the 98-clip training set. Dropout ($p = 0.40$) and AdamW weight decay ($\lambda = 3 \cdot 10^{-4}$) regularise training. [2]

C. Training

Five GRU models are trained using stratified subsampling (80% per class per fold) with out-of-bag (OOB) early stopping (patience = 25 epochs) and five noisy test-time augmentation (TTA) passes at inference. Two additional learners (a scikit-learn RandomForestClassifier (500 trees) and a HistGradientBoostingClassifier (150 iterations) [3] [4]) are trained on the 71-dim global feature vector. Blend weights (wGRU, wRF, wHGB) are determined by grid search over out-of-fold probability estimates, maximising micro-F1. The RF and HGB complement the GRU because they partition tabular feature space directly rather than learning it implicitly from sequences.

III. TEMPORAL ACTION SEGMENTATION APPROACH

A. Feature Engineering

Only proprioceptive signals were used as input, since RGB images require an expensive visual backbone and early experiments showed the proprioceptive signal alone carries sufficient discriminative information for this task. The raw proprioceptive vector per frame consists of 37 dimensions (7 pose, 6 velocity, 7 joint positions, 7 joint velocities, 2 gripper positions, 2 gripper velocities, 6 wrench). To capture temporal dynamics explicitly, first-order differences and second-order differences were appended, giving a 111-dim feature vector per frame. These derivatives encode instantaneous velocity and acceleration of the robot state, which are the primary signals for detecting action phase transitions.

B. Model Architecture

A 3-layer bidirectional GRU (BiGRU) [5] with hidden dimension 128 per direction (256 total) was used. The BiGRU processes the full session sequence and at each timestep combines information from both past and future frames, which is important for distinguishing actions near boundaries. A LayerNorm layer precedes the linear classification head. The total parameter count is approximately 780k, which was found to be appropriate for a training set of 98 sessions - substantially larger models (e.g. MS-TCN [6] at 5M parameters) overfit severely and produced degraded performance.

C. Training

The model was trained with class-weighted cross-entropy loss. Without class weighting, the dominant class biases the model toward predicting background. Weights were set inversely proportional to class frequency. The Adam optimiser [7] was used with an initial learning rate of $5 \cdot 10^{-4}$, weight decay $1 \cdot 10^{-4}$, and a cosine annealing schedule decaying to $lr/50$. Mild Gaussian noise ($\sigma = 0.02$) was added to features each epoch as data augmentation. Training was run for 20 epochs, determined by monitoring validation F1@0.50 - longer runs (30-50 epochs) caused overfitting, significantly decreasing F1@0.50 result.

IV. RESULTS

A. Quantitative Evaluation

The quantitative performance of the proposed methods is shown in Tables I and II for Action Recognition and Temporal Action Segmentation respectively. It is important to mention that only some of the versions of the model are included.

TABLE I
ACTION RECOGNITION PERFORMANCE

Version	Configuration changes	F1 (%)
v3.0.1	BiGRU, mean + max pooling, 30 epochs	74.38
v4.0	1D temporal convolutions, self-attention pooling	68.13
v9.3	Dual-branch + bagging ensemble	86.88
v9.6	Per-frame repair, RF+HGB added, OOF blend search	89.38
v9.8	Terminal grip width; gripper sequence deltas reverted	90.00

TABLE II
TEMPORAL ACTION SEGMENTATION PERFORMANCE

Version	Configuration	F1@0.5 (%)	Accuracy (%)
v2.0.30.3	2-layer BiGRU, 30 epochs	70.71	73.12
v3.0	MS-TCN	40.66	n/a
v4.0	3-layer BiGRU, 50 epochs	72.35	84.50
v4.0.30	3-layer BiGRU, 30 epochs	77.30	84.73
v4.0.20	3-layer BiGRU, 20 epochs	86.36	87.54

B. Qualitative Evaluation

The qualitative performance of the proposed methods is shown in Figures 1 and 2 for Action Recognition and Temporal Action Segmentation respectively.

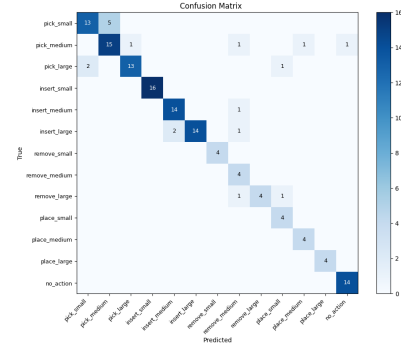


Fig. 1. Confusion matrix for validation set - AR

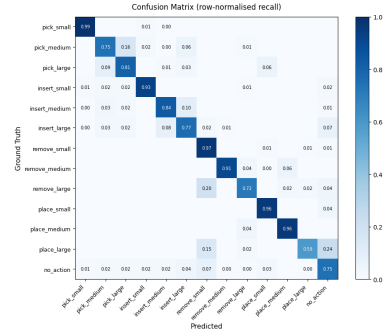


Fig. 2. Normalised confusion matrix for validation set - TAS

V. CONCLUSION

This project has shown that there is no universally applicable solution - every machine learning problem requires unique approach. Despite the fact that bigger models, more epochs or parameters can sound like the better solution at first, it does not have to be the case.

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